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Multi-Fidelity Dynamic Line Rating Fusion for System Load Margin Enhancement with Large-Scale Offshore Wind Generations

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Abstract—This paper proposes a multi-fidelity dynamic line rating (DLR) fusion method for power system load margin enhancement with large-scale offshore wind integration. First, multi-fidelity DLR data from diverse sensors are fused using multi-fidelity deep sigma point processes (MFDSPP), a probabilistic nonlinear learning model that integrates heterogeneous data sources to estimate the minimum line rating and quantify its uncertainty. Conformal prediction (CP), a distribution-free uncertainty calibration technique that refines prediction intervals using empirical residuals, is developed to refine the uncertainty intervals of DLRs. Then, continuation power flow (CPF) analysis is used to calculate the system's maximum load margin for numerous $N - k$ contingencies. Sampling these scenarios generates a large dataset to train the deep sigma point processes (DSPP) model, establishing a direct nonlinear relationship between operational data and load margin. Finally, DLR-based line capacity constraints and DSPP-derived load-margin constraints are integrated into the real-time control framework to improve system stability. Case studies on the Northeast Power Coordinating Council (NPCC) system, incorporating real DLR sensor measurements, authentic $N - k$ contingency datasets, and four real large-scale offshore wind farms, demonstrate that the proposed method effectively maintains a load margin greater than 10% of the base load, unlocks an additional 2,189.7 MWh of clean energy, reduces operating costs by approximately 2.34%, and achieves more than a 2,000× speedup compared with classical CPF-based sensitivity methods.

Index Terms—Continuation power flow, dynamic line rating, load margin enhancement, multi-fidelity data fusion, $N - k$ contingencies, offshore wind farm, voltage stability.

NOMENCLATURE

Notations

α Desired miscoverage level for CP, $\alpha \in (0, 1)$

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β_{reg}	Regularization weight for KL-divergence terms, $\beta_{\text{reg}} \geq 0$
δ	Dirac delta function
η	Confidence level coefficient in the probabilistic model
$\lambda/\underline{\lambda}$	Load margin/minimum λ required to maintain safe operation of the power system
μ/σ	Mean/standard deviation estimated from DSPP or MFDSPP
$\mu_{g_w}/\sigma_{g_w}^2$	Posterior mean/variance of latent variable g_w
μ_f/σ_f^2	Mean/variance function of the DSPP output evaluated at a sigma-point
$\mu_{y_H}/\sigma_{y_H}^2$	Mean/variance of the estimated DLR by MFDSPP
$\mu_{\lambda}/\sigma_{\lambda}^2$	Mean/variance of the estimated λ by DSPP
ρ	Activation function
σ_w^2	Observation-noise variance
$\xi_w^{(s)}$	s -th learnable sigma-point for latent dimension w
$\omega^{(s)}$	Weight of s -th sigma-point, $\sum_s \omega^{(s)} = 1$
$a_{\text{SG}}/b_{\text{SG}}/c_{\text{SG}}$	Quadratic/linear/fixed cost coefficient of SG
C_g	Connection matrix that maps SG and RES injections to buses
$C^{1-\alpha}$	CP interval with coverage level $1 - \alpha$
$\mathcal{D}$	Set of samples $\mathcal{D} = \{\mathbf{x}, y\}$
$\mathcal{D}_t$	Set of calibration $\mathcal{D}_t = \{\mathbf{x}_t, y_t\}$ with size m
d_{RES}	Cost constant of $p_{\text{RES},t}$ curtailment
$\mathcal{F}$	Nonlinear function mapping $\mathbf{x}_L$ and $\mathbf{x}_H$ to the fused DLR estimate
$\mathcal{F}_L/\mathcal{F}_H$	Low/high-fidelity network in $\mathcal{F}(\cdot)$
$\mathcal{F}_{\lambda}/\mathbf{J}_{\lambda}/\mathbf{H}_{\lambda}$	Value/Jacobian/Hessian matrix of the λ chance constraint function
$\mathbf{f}_L^{(l)}/\mathbf{f}_H^{(l)}$	Feature representations of $\mathcal{F}_L/\mathcal{F}_H$ at layer l
$\mathcal{G}_p/\mathcal{G}_q$	Active/reactive power balance equation
g_w	Output of the w -th latent function channel
$\tilde{g}^{(s)}$	Latent-variable to s -th sigma-point
$\inf\{\cdot\}$	Infimum, i.e. the greatest lower bound of all q satisfying the constraint
KL	Kullback–Leibler divergence
K	Number of layers in $\mathcal{F}_L$
$\mathcal{L}_{\text{DSPP}}$	Regularized log-likelihood objective for DSPP training
$\mathcal{N}$	Normal distribution function
$\mathcal{P}_{\text{bus}}/\mathcal{Q}_{\text{bus}}$	Net active/reactive power injections at buses

	computed from the AC network equations
$p_{\text{DSPP}}(y \mathbf{x})$	Predictive distribution of DSPP for input $\mathbf{x}$
$\mathbf{p}_g/\mathbf{q}_g$	Active/reactive power injections from all SGs and RESs aggregated at buses
$\mathbf{p}_d/\mathbf{q}_d$	Active/reactive power demands
$\mathbf{p}_{\text{SG},t}/\mathbf{p}_{\text{RES},t}$	Active power generation of SG/RES
$\bar{\mathbf{p}}_{\text{SG}}/\underline{\mathbf{p}}_{\text{SG}}$	Upper/lower limits of $\mathbf{p}_{\text{SG},t}$
$\hat{\mathbf{p}}_{\text{RES}}$	Maximum active power output of $\mathbf{p}_{\text{RES},t}$ under current weather conditions
$\tilde{\mathbf{p}}_{\text{SG},t}$	Active power generation of SG after applying the specified increments in CPF
$\tilde{\mathbf{p}}_{d,t}/\tilde{\mathbf{q}}_{d,t}$	Active/reactive power demands after applying the specified increments in CPF
$\mathbf{p}_{l,t}$	Active power flow of transmission lines
$\bar{\mathbf{p}}_l^{\text{SLR}}/\underline{\mathbf{p}}_l^{\text{DLR}}$	Maximum active power flow limit of $\mathbf{p}_{l,t}$ based on SLR/DLR
$\mathbf{q}_{\text{SG},t}/\mathbf{q}_{\text{RES},t}$	Reactive power generation of SG/RES
$\bar{\mathbf{q}}_{\text{SG}}/\underline{\mathbf{q}}_{\text{SG}}$	Upper/lower limits of $\mathbf{q}_{\text{SG},t}$
$\bar{\mathbf{q}}_{\text{RES}}/\underline{\mathbf{q}}_{\text{RES}}$	Upper/lower limits of $\mathbf{q}_{\text{RES},t}$
$q(g_w \mathbf{x})$	Gaussian variational posterior distribution of latent variable g_w with input data $\mathbf{x}$
$q^{1-\alpha}$	Empirical $(1 - \alpha)$ -quantile
$q_{y_H}^{1-\alpha}/q_{\lambda}^{1-\alpha}$	$q^{1-\alpha}$ for line capacity/ λ constraint
$\mathbf{v}_t/\boldsymbol{\theta}_t$	Bus voltage magnitude/phase angle
$\bar{\mathbf{v}}/\underline{\mathbf{v}}$	Upper/lower limits of $\mathbf{v}_t$
S	Number of sigma-points used in DSPP
s_{RES}	Rated apparent power of RESs
t	Time instant
u_ℓ	Inducing variables at layer ℓ
$\mathbf{W}_L^{(l)}/\mathbf{b}_L^{(l)}$	Weights/biases of $\mathcal{F}_L$ at layer l
W	Number of latent dimensions
$\mathbf{x}/\mathbf{y}$	Input/output feature vector
$\mathbf{x}_\lambda$	n -dim input variables for λ estimation
$\mathbf{x}_L/\mathbf{x}_H$	Low/high-fidelity DLR measurements
y_L/y_H	Low/high-fidelity DLR estimates obtained from $\mathcal{F}_L/\mathcal{F}_H$

Abbreviations

(AC)OPF	(Alternating current) optimal power flow
CP	Conformal prediction
CPF	Continuation power flow
DBN	Deep belief network
DGP	Deep Gaussian process
DKL	Deep kernel learning
DLR	Dynamic line rating
DNN	Deep neural network
DSPP	Deep sigma point processes
GP	Gaussian process
MAE	Mean absolute error
MFDKL	Multi-fidelity deep kernel learning
MFDSPP	Multi-fidelity deep sigma point processes
MFNN	Multi-fidelity neural network
NPCC	Northeast Power Coordinating Council
PDIPM	Primal-dual interior-point method
R^2	Coefficient of determination
RES	Renewable energy source
RMSE	Root mean square deviation

SG	Synchronous generator
SLR	Static line rating

I. INTRODUCTION

THE continuous growth of stochastic RES and load imposes significant challenges on existing power systems, particularly concerning transmission line capacities and load margins [1]. Transmission line capacity is the maximum power a line can carry without exceeding thermal limits, and current practice generally relies on SLRs based on conservative worst-case ambient assumptions. However, SLRs often underutilize existing infrastructure [2]. Meanwhile, the increased variability and uncertainty of RES lead to stochastic power flows, frequent congestion, elevated risks of transmission line violations, and increased RES curtailment [3]. Load margin measures how much additional load a power system can serve before reaching voltage stability limits, and larger margins indicate that the system is farther from voltage collapse and more resilient to demand spikes or generator outages [4]. **Under high-load and high-ambient-temperature conditions, transmission lines approach their thermal limits and available load margins shrink, making the goals of maximizing power transfer and preserving load margin potentially conflicting.** Therefore, it is necessary to develop effective control methods that enhance line utilization while maintaining adequate load margins.

DLR improves capacity estimates by leveraging indirect meteorological data or direct sensor measurements, often yielding higher limits than conservative SLRs [5]. However, under severely adverse weather, DLR may fall below SLR, and adjusting the line rating accordingly can help prevent unnecessary failures. DLR is calculated based on measured or forecasted weather conditions and is typically higher than SLR [6]. According to [7] and [8], adopting DLR can improve network reliability, support higher levels of RES integration, and unlock significant economic potential. A variety of power system control methods incorporating DLR have been proposed [9], [10]. In [11], an accurate electrothermal coordination model is formulated by treating overhead line temperature as a decision variable within the ACOPF framework and incorporating the heat-balance equations of the lines as equality constraints. [12] presents an ACOPF that integrates weather-based conductor thermal limits for high-renewable systems, solves it with a primal and dual interior point solver. [3] introduces a data-driven, distributionally robust OPF with DLR that embeds risk penalties for load shedding, wind curtailment, and line overload, and hedges against uncertainty using a moment-based correlation set and a Wasserstein-distance constraint. Furthermore, DLR has been considered in transmission expansion planning [13] and the mitigation of cascading failure risk [14], demonstrating its applications in various optimization, planning, and fault mitigation problems. These applications have demonstrated significant benefits in terms of reliability, safety, and economic efficiency. However, most existing studies rely on meteorological data-based DLR, which can be inaccurate and deviate substantially from actual values. Moreover, research on fusing multiple DLR data sources to enhance rating accuracy is lacking.

Load margin is a key operational metric used by control centers to assess voltage stability and has been extensively studied, with both model-based and data-driven methods [15]. Model-based methods obtain the load margin distribution through extensive CPF analyses for all contingencies, which is time-consuming, especially in large power systems with numerous contingencies [16]. To obtain probabilistic load margin assessments, the Monte Carlo simulation method is employed, which further increases the evaluation burden [17]. The above studies focus solely on load margin estimation, and real-time control requires repeated evaluations and sensitivity analysis of control variables, which makes the computation prohibitive. With the widespread deployment of measurement devices, data-driven methods for assessing load margin have been developed [15]. For example, [18] proposes an online voltage stability margin monitoring method that uses local regression and an adaptive database to capture the relationship between voltage stability margin and reactive power reserve under changing system conditions. [19] proposed a physics-guided neural network to calculate power system load margins, regularized by an auxiliary process that reconstructs the power flow equations at the stability threshold defining the margin. [20] presents a cost-effective, data-driven method for probabilistic load margin assessment by estimating wind speed distributions with a kernel density estimator and integrating them into a vine copula framework. In [21], a probabilistic transferable deep kernel emulator is proposed to uncover the hidden relationship between uncertain sources and load margin for probabilistic load margin assessment. However, these assessment methods are not integrated with control and thus cannot demonstrate their benefits in improving system voltage stability. In addition, most existing studies on load margin enhancement focus on installing flexible alternating current transmission system (FACTS) devices rather than on real-time control [22], [23].

Uncertainty quantification is critical in safety-critical domains like power system assessment and control. GPs offer a Bayesian nonparametric approach that yields well-calibrated uncertainty estimates when their model assumptions (e.g., smoothness or periodicity) hold. However, in real-time applications, the assumptions of GPs may fail as data streams in and distribution shifts occur, invalidating the original GP framework. To address this limitation, GPs can be combined with CP, a data-driven uncertainty calibration technique that adjusts prediction intervals using observed prediction errors and does not rely on distributional assumptions, thereby generating prediction intervals with guaranteed coverage for any specified miscoverage rate. Compared with parametric or likelihood-based intervals, CP remains well-calibrated under model misspecification because the interval width is calibrated from empirical residuals rather than assumed error distributions [24], [25].

To address the above research gaps, this paper proposes a multi-fidelity DLR fusion-based load margin enhancement method with large-scale offshore wind integration. The main contributions are summarized as follows:

- 1) To handle both low-fidelity meteorological-based DLR and high-fidelity sensor-based DLR data, we propose

a MFDSPP-based DLR estimation framework, which is a probabilistic learning model designed to fuse heterogeneous data sources while capturing nonlinear relationships and estimation uncertainty. We also apply CP, a distribution-free post-processing technique that calibrates predictive uncertainty using empirical residuals, to GP estimates to refine the DLR uncertainty intervals.

- 2) Conventional CPF is time-consuming, especially in scenarios with many $N - k$ contingencies. We propose a data-driven surrogate model based on DSPP to rapidly assess load margin and integrate CP to quantify and calibrate uncertainty. In addition, our data-driven approach directly establishes a nonlinear relationship between control variables and load margin, enabling fast computation of the Jacobian and Hessian of the load margin with respect to control variables.
- 3) A real-time control framework is proposed that integrates both DLRs and load margin constraints. In this framework, the fused DLR data are used to formulate transmission line capacity constraints, while the DSPP surrogate model constructs chance constraints on load margin. Case studies demonstrate the proposed method's effectiveness in increasing load margin, improving RES utilization, and mitigating potential risks.

The rest of the paper is organized as follows. Section II presents the traditional real-time load margin enhancement framework. Section III presents the proposed real-time control framework that incorporates DLRs and load margin chance constraint. Results are presented and analyzed in Section IV, and finally, Section V concludes the paper.

II. PROBLEM STATEMENT

This section introduces the real-time load margin enhancement formulation considering SLR constraints, which motivates the proposed framework of DLR-based solutions.

A. Objective Function

The objective function is to minimize the SG generation cost $\mathcal{M}_{SG,t}$ and RES curtailment cost $\mathcal{M}_{RES,t}$:

$$\begin{aligned} \min \quad & \mathcal{M}_{SG,t}(\mathbf{p}_{SG,t}) + \mathcal{M}_{RES,t}(\mathbf{p}_{RES,t}) \\ \mathcal{M}_{SG,t}(\mathbf{p}_{SG,t}) = & \sum \mathbf{a}_{SG} \mathbf{p}_{SG,t}^2 + \mathbf{b}_{SG} \mathbf{p}_{SG,t} + \mathbf{c}_{SG} \quad (1) \\ \mathcal{M}_{RES,t}(\mathbf{p}_{RES,t}) = & \sum \mathbf{d}_{RES} (\hat{\mathbf{p}}_{RES,t} - \mathbf{p}_{RES,t}) \end{aligned}$$

B. AC Power Balance Constraints

The power balance constraints are formulated at the nodal level, where active and reactive power balance equations are enforced at each bus using the standard AC power flow model. Specifically, the active power balance equation $\mathcal{G}_p$ and reactive power balance equation $\mathcal{G}_q$ are given by:

$$\mathcal{G}_p(\boldsymbol{\theta}, \mathbf{v}, \mathbf{p}_g, \mathbf{p}_d) = \mathcal{P}_{bus}(\boldsymbol{\theta}, \mathbf{v}) + \mathbf{p}_d - \mathbf{C}_g \mathbf{p}_g = \mathbf{0} \quad (2)$$

$$\mathcal{G}_q(\boldsymbol{\theta}, \mathbf{v}, \mathbf{q}_g, \mathbf{q}_d) = \mathcal{Q}_{bus}(\boldsymbol{\theta}, \mathbf{v}) + \mathbf{q}_d - \mathbf{C}_g \mathbf{q}_g = \mathbf{0} \quad (3)$$

These equations represent the net power balance at each bus, in which power exchanges with neighboring buses are implicitly

captured through the nonlinear AC network equations. As a result, transmission losses and network coupling effects are fully accounted for.

C. SLR-Based Transmission Line Capacity Constraints

$$|p_{l,t}| \leq \bar{p}_l^{\text{SLR}} \quad (4)$$

In this paper, (4) is enforced as a hard thermal limit in the ACOPF formulation, meaning that short-term overloads or emergency ratings are not allowed. This modeling choice reflects practical operational considerations, as temporary overloading may lead to irreversible thermal degradation of transmission lines, and the associated lifetime and degradation costs are difficult to quantify accurately. Moreover, adopting the same hard-constraint assumption for both SLR- and DLR-based formulations ensures a fair and consistent comparison of their operational performance.

D. Variable Limits

Variable limits include bus voltage magnitude constraints, SG active and reactive power output constraints, and RES generator active and reactive power output constraints:

$$\underline{v} \leq v_t \leq \bar{v} \quad (5)$$

$$\underline{p}_{\text{SG}} \leq p_{\text{SG},t} \leq \bar{p}_{\text{SG}}, \quad \underline{q}_{\text{SG}} \leq q_{\text{SG},t} \leq \bar{q}_{\text{SG}} \quad (6)$$

$$0 \leq p_{\text{RES},t} \leq \hat{p}_{\text{RES},t} \leq s_{\text{RES}}, \quad \underline{q}_{\text{RES}} \leq q_{\text{RES},t} \leq \bar{q}_{\text{RES}} \quad (7)$$

$$\bar{q}_{\text{RES}}/\underline{q}_{\text{RES}} = +/\sqrt{(s_{\text{RES}})^2 - (p_{\text{RES}})^2} \quad (8)$$

The limits in (5) - (8) are enforced as hard operational bounds in the real-time optimization, and transient or short-term exceedances are not explicitly modeled. This choice reflects standard operational practice, as relaxing these limits may violate network security or stability requirements and introduce additional degrees of freedom that are not physically realizable under normal operating conditions. Moreover, enforcing consistent hard limits avoids implicitly favoring certain control strategies and ensures a fair comparison among different formulations.

E. Load Margin Constraint

The load margin is obtained via CPF, which traces a continuum of power flow solutions from a base load to the system's steady-state voltage stability limit (critical or nose point). In particular, CPF determines the maximum loading factor λ by solving the following parameterized power flow problem [26]:

$$\mathcal{G}_p(\theta, \mathbf{v}, \tilde{\mathbf{p}}_g, \tilde{\mathbf{p}}_d) = \mathcal{G}_q(\theta, \mathbf{v}, \tilde{\mathbf{q}}_g, \tilde{\mathbf{q}}_d) = 0 \quad (9)$$

$$\tilde{\mathbf{p}}_{\text{SG},t} = \mathbf{p}_{\text{SG},t} + \lambda \Delta \mathbf{p}_{\text{SG},t} \quad (10)$$

$$\tilde{\mathbf{p}}_{d,t} = \mathbf{p}_{d,t} + \lambda \Delta \mathbf{p}_{d,t}, \quad \tilde{\mathbf{q}}_{d,t} = \mathbf{q}_{d,t} + \lambda \Delta \mathbf{q}_{d,t} \quad (11)$$

where $\mathbf{p}_{\text{RES},t}$ is constrained by weather conditions and is therefore not increased. By progressively increasing the loading parameter λ , CPF uses iterative predictor and corrector steps to determine the critical point, which is identified when

those steps no longer produce any further increase in load, indicating that the system has reached its maximum loading limit [26]. The proposed framework assumes that the system operates at a feasible base operating point with a convergent AC power flow solution, which is a standard prerequisite for CPF-based voltage stability assessment. It is worth noting that load margin is not a directly measurable quantity in real power systems, as it corresponds to the maximum loading point prior to voltage collapse, which cannot be experimentally reached for safety reasons. As a result, CPF has been widely adopted as the standard physics-based tool for steady-state voltage stability assessment and is commonly used in both academic studies and industry-grade software. In this work, CPF is employed as the physical benchmark for margin computation under realistic operating conditions, including real DLR sensor measurements, authentic $N - k$ contingency sets, and real offshore wind farm data.

CPF is time-consuming, especially with many contingencies, because it needs to compute the load margin for all contingencies and then select the smallest value. Finally, the load margin constraint is defined as:

$$\lambda \geq \underline{\lambda} \quad (12)$$

where λ is a dimensionless loading parameter that scales the specified active and reactive power increment directions in the CPF. Because (12) requires executing (9) - (11) to verify compliance, embedding this process in a real-time control framework makes online computation infeasible.

III. LOAD MARGIN ENHANCEMENT FRAMEWORK CONSIDERING MULTI-FIDELITY DLR FUSION

A. Multi-Fidelity DLR Fusion

A transmission line between two buses consists of multiple conductor spans, defined as the segments of conductor between adjacent towers, and its overall thermal capacity is determined by the span with the lowest rating. SLRs cannot capture real-time, span-specific variations in conductor conditions and thus often lead to overly conservative capacity estimates. Therefore, DLR has been proposed with two common calculation methods. The first relies on indirect meteorological data, as defined in IEEE Std. 738 [27] and CIGRE SC.B2 WG 22.12 [28], both of which produce identical results [29], [30]. The second uses sensors attached directly to conductors or towers, or placed on the ground to measure variables such as conductor temperature, sag, line tension, or clearance from the ground [31].

Existing methods typically use a single DLR calculation technique. However, relying solely on meteorological-based estimates is insufficiently accurate for real networks, and installing sensors on every span between two buses is prohibitively expensive. To solve this, we propose a multi-fidelity DLR fusion approach that combines low-fidelity meteorological-based DLRs for all spans with high-fidelity sensor-based DLRs on selected critical spans to estimate the line's overall DLR. In practical DLR deployments, sensors are installed on transmission lines identified as critical based on their operational importance, such as high loading levels,

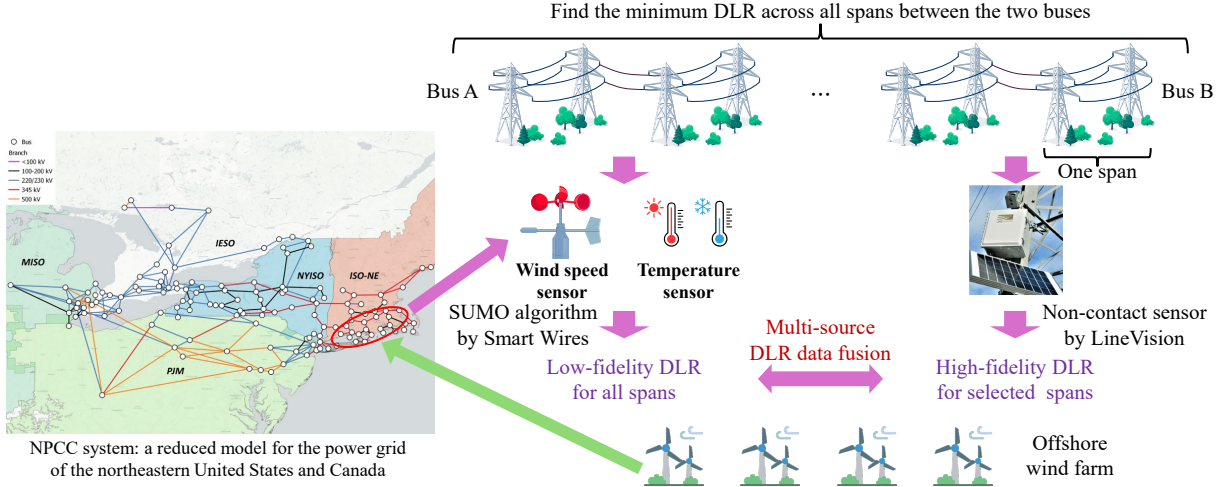


Fig. 1. Multi-Fidelity DLR Data Fusion.

significant power flow variability, proximity to large renewable generation, and frequent binding constraints in contingency analysis. Fig. 1 shows the multi-fidelity DLR data fusion architecture: low-fidelity estimates are calculated from meteorological data, while high-fidelity measurements come from non-contact sensors, such as those from LineVision, a commercial provider of DLR monitoring solutions.

To implement multi-fidelity DLR fusion, we develop a MFDSP network that jointly ingests low-fidelity and high-fidelity DLR measurements. Multi-fidelity modeling fundamentally relies on uncovering and exploiting the relationship between these fidelity levels. The generalized autoregressive scheme that captures nonlinear correlations is [32], [33]:

$$y_H = \mathcal{F}(x_L, x_H) \quad (13)$$

To leverage the comprehensive coverage of low-fidelity data across all spans and the higher accuracy of high-fidelity measurements on selected spans, we decompose $\mathcal{F}$ into two components: a low-fidelity network $\mathcal{F}_L$ and a high-fidelity network $\mathcal{F}_H$.

$$y_L = \mathcal{F}_L(\mathbf{f}_L^{(0)}), \quad \mathbf{f}_L^{(0)} = \mathbf{x}_L, \quad \mathbf{f}_L^{(K)} = y_L \quad (14)$$

$$\mathbf{f}_L^{(l)} = \rho(\mathbf{W}_L^{(l)} \mathbf{f}_L^{(l-1)} + \mathbf{b}_L^{(l)}), \quad l = 1, \dots, K \quad (15)$$

$$y_H = \mathcal{F}_H(\mathbf{f}_H^{(0)}), \quad \mathbf{f}_H^{(0)} = [\mathbf{x}_H, y_L, \mathbf{f}_L^{(K-1)}]^\top \quad (16)$$

The low-fidelity mapping $\mathcal{F}_L$ is implemented as a DNN whose hierarchical layers successively abstract span-level DLR patterns into richer feature representations. Moreover, rather than relying only on the final output y_L , the fusion stage also ingests activations from the penultimate layer $\mathbf{f}_L^{(K-1)}$, capturing both high-level and intermediate abstractions to provide a more comprehensive foundation across multiple scales for the high-fidelity network.

Because DLR is influenced by random factors such as weather conditions, the high-fidelity mapping $\mathcal{F}_H$ is implemented using the DSPP, a fully parametric alternative for DGPs by replacing the intractable integral over each latent

layer with a finite, learnable Gaussian mixture. By casting each layer's continuous integral as a finite weighted sum over sigma-points, DSPPs preserve the compositional expressivity of deep kernels while enabling straightforward maximum-likelihood training without Monte Carlo sampling or variational bounds. Concretely, for input $\mathbf{x} \triangleq \mathbf{f}_H^{(0)}$ and w latent channel, the Gaussian variational posterior $q(g_w|\mathbf{x})$ is:

$$q(g_w|\mathbf{x}) = \mathcal{N}(g_w|\mu_{g_w}(\mathbf{x}), \sigma_{g_w}^2(\mathbf{x})) \quad (17)$$

Rather than integrating over this continuous distribution, DSPP employs S learnable “sigma-points” $\{\xi_w^{(s)}\}_{s=1}^S$ and weights $\{\omega^{(s)}\}$ (with $\sum_s \omega^{(s)} = 1$) to form the finite mixture:

$$q(g_w|\mathbf{x}) = \sum_{s=1}^S \omega^{(s)} \delta(g_w - [\mu_{g_w}(\mathbf{x}) + \xi_w^{(s)} \sigma_{g_w}(\mathbf{x})]) \quad (18)$$

This discretization yields a closed-form predictive distribution

$$p_{\text{DSPP}}(y|\mathbf{x}) = \sum_{s=1}^S \omega^{(s)} \mathcal{N}(y|\mu_f(\tilde{\mathbf{g}}^{(s)}), \sigma_f^2(\tilde{\mathbf{g}}^{(s)}) + \sigma_{\text{obs}}^2) \quad (19)$$

where $\tilde{\mathbf{g}}^{(s)} = (\mu_{g_1}(\mathbf{x}) + \xi_1^{(s)} \sigma_{g_1}(\mathbf{x}), \dots, \mu_{g_W}(\mathbf{x}) + \xi_W^{(s)} \sigma_{g_W}(\mathbf{x}))$ collects the S sigma-point vectors across all W latent dimensions. The model is trained by maximizing the regularized log-likelihood:

$$\mathcal{L}_{\text{DSPP}} = \sum_{(\mathbf{x}, y) \in \mathcal{D}} \log p_{\text{DSPP}}(y|\mathbf{x}) - \beta_{\text{reg}} \sum_{\ell} \text{KL}(q(u_{\ell})|p(u_{\ell})) \quad (20)$$

The MFDSP architecture is visualized in Fig. 2. Based on this, the proposed MFDSP framework not only fuses low- and high-fidelity DLR measurements and their intermediate feature representations, but also yields a Gaussian uncertainty estimate. To guarantee sufficient safety margins, we employ a one-sided confidence bound. Accordingly, (4) is revised as:

$$|p_{l,t}| \leq \bar{p}_l^{\text{DLR}} = \mu_{y_H} - \eta \sigma_{y_H} \quad (21)$$

where the confidence level η can be determined using the cumulative distribution function of the normal distribution.

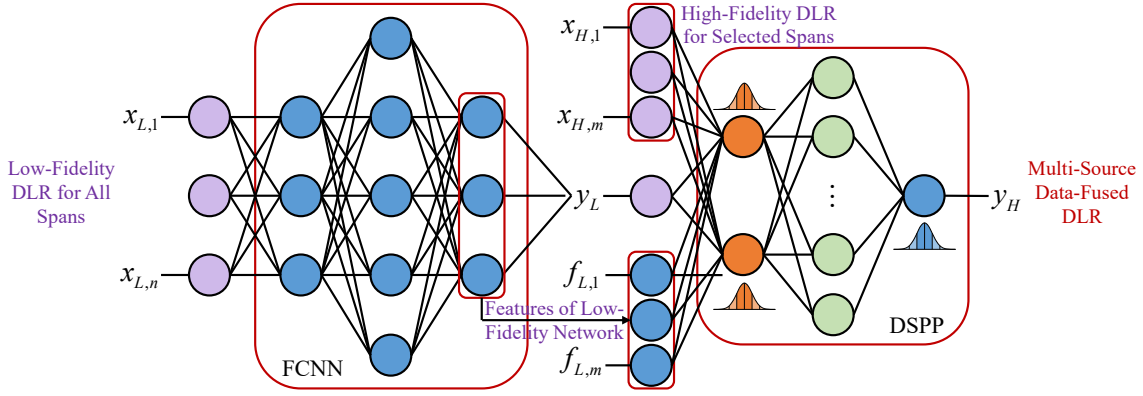


Fig. 2. The structure of the MFDSPP.

B. Conformal Prediction

While DSPP or MFDSPP provides well-calibrated uncertainty estimates under assumptions like smoothness or periodicity, these assumptions often fail in real-time, streaming settings where distribution shifts occur. CP offers a distribution-free, post-processing framework that guarantees prediction intervals with any specified miscoverage rate as long as data are exchangeable [24], [25]. By applying CP to DSPP outputs, we obtain refined uncertainty intervals that remain valid even when the original model assumptions are violated. Given a calibration dataset $\mathcal{D}_t = \{\mathbf{x}_t, y_t\}$, we first compute a standardized-residual score for each calibration pair by:

$$s_t(\mathbf{x}, y) = \frac{|y - \mu(\mathbf{x})|}{\sigma(\mathbf{x})} \quad (22)$$

The empirical $(1 - \alpha)$ -quantile $q^{1-\alpha}$ of scores $\{s_t(\mathbf{x}_j, y_j)\}_{j=1}^m$ on the $\mathcal{D}_t$ is:

$$q^{1-\alpha} = \inf \left\{ q : \frac{1}{m} \sum_{j=1}^m \mathbf{1}(s_t(\mathbf{x}_j, y_j) \leq q) \geq 1 - \alpha \right\} \quad (23)$$

This score yields conformal intervals of the form

$$C^{1-\alpha}(\mathbf{x}) = \mu(\mathbf{x}) \pm q^{1-\alpha} \sigma(\mathbf{x}) \quad (24)$$

which remains valid for heteroscedastic and mixture predictive models. The resulting CP set satisfies the coverage guarantee:

$$\mathbb{P}(y \in C^{1-\alpha}(\mathbf{x})) \geq 1 - \alpha \quad (25)$$

Accordingly, the line capacity constraint (21) is updated as:

$$|p_{l,t}| \leq \bar{p}_l^{\text{DLR}} = \mu_{y_H} - q_{y_H}^{1-\alpha} \sigma_{y_H} \quad (26)$$

C. DSPP-Based Load Margin Chance Constraint

Traditionally, enforcing the load-margin constraint (12) requires running the CPF procedure defined by (9) - (11). In a real-time control loop, this means executing CPF at every iteration to check constraint satisfaction, which is too slow for online use. To address this, we introduce a DSPP surrogate model that replaces the iterative CPF solver with a single feed-forward inference pass, thereby greatly reducing computation time. The inputs to the DSPP are various real-time power system measurements, namely $\mathbf{x}_\lambda = [\mathbf{p}_{\text{RES}}, \mathbf{p}_{\text{SG}}, \mathbf{p}_d, \mathbf{q}_d]^\top$.

The DSPP outputs both a predictive mean μ_λ and standard deviation σ_λ for the minimum load margin:

$$(\mu_\lambda(\mathbf{x}_\lambda), \sigma_\lambda(\mathbf{x}_\lambda)) = \mathcal{F}_H(\mathbf{x}_\lambda) \quad (27)$$

We then model the load margin λ as a Gaussian random variable:

$$\lambda \sim (\mu_\lambda(\mathbf{x}_\lambda), \sigma_\lambda^2(\mathbf{x}_\lambda)) \quad (28)$$

During training, the DSPP is fit to CPF-computed load margins across all contingencies, but in real-time it replaces those costly iterations with a single, efficient inference step. Furthermore, since the estimated λ follows a normal distribution, we can formulate a chance constraint on λ , transforming (12) into:

$$\mu_\lambda - \eta \sigma_\lambda \geq \underline{\lambda} \quad (29)$$

Similarly, we apply the CP described in Section III-B to calibrate the DSPP outputs, yielding the following chance constraint on load margin:

$$\mu_\lambda - q_\lambda^{1-\alpha} \sigma_\lambda \geq \underline{\lambda} \quad (30)$$

In the current implementation, the quantile $q_\lambda^{1-\alpha}$ is computed offline from a held-out calibration set and kept fixed during deployment. If significant distribution shift occurs, the calibration quantile may become miscalibrated; in such cases, online or rolling calibration strategies that periodically refresh the quantile using newly available samples can be employed.

D. Chance-Constrained Load Margin Enhancement Framework Considering Multi-Fidelity DLR Fusion

Based on Section II, a novel real-time chance-constrained load-margin enhancement framework is proposed that incorporates multi-fidelity DLR fusion. The real-time aspect corresponds to a rolling operational formulation with an hourly decision-making horizon, which can be extended to finer temporal resolutions (e.g., 15-minute or 5-minute intervals). The objective function is given by (1), the power flow balance constraints by (2) and (3), the variable limits by (5) - (8), the DLR-based transmission line capacity constraint by (26), and the load margin chance constraint by (30). It should be noted that we can define the function $\mathcal{F}_\lambda = \underline{\lambda} - \mu_\lambda + q_\lambda^{1-\alpha} \sigma_\lambda$, and obtain its Jacobian $\mathbf{J}_\lambda = d\mathcal{F}_\lambda/d\mathbf{x}_\lambda$ and Hessian

Algorithm 1: Chance-constrained load margin enhancement considering multi-fidelity DLR fusion.

Input: $x_L, x_H, \hat{p}_{\text{RES}}, p_d, q_d, \zeta = 0$

Output: Real-time control strategies p_{RES} and p_{SG}

- 1 **Offline Stage:**
 - 2 Define the feasible range of x_λ , specify the set of contingencies, perform CPF for each scenario, and calculate the minimum load margin in each case.
 - 3 Train the DSPP to develop a surrogate model for load margin estimation.
 - 4 Train the MFDSPP using historical data x_L and x_H .
 - 5 **Online Stage:**
 - 6 MFDSPP processes real-time measurements x_L and x_H to produce y_H . Its uncertainty intervals are refined via CP and used to formulate the transmission line capacity constraint in (26).
 - 7 The optimization problem is defined with the objective function in (1) and is subject to constraints (2), (3), (5) to (8), (26) and (30). It is solved by PDIPM.
 - 8 **while** Not converge **or** below max iterations **do**
 - 9 Build input $x_\lambda^\zeta = [p_{\text{RES}}^\zeta, p_{\text{SG}}^\zeta, p_d^\zeta, q_d^\zeta]^\top$ for DSPP.
 - 10 Compute the value, Jacobian, and Hessian of (1), (2), (3), (5) to (8), and (26).
 - 11 Compute the value, Jacobian, and Hessian of (30) using built-in automatic differentiation functions.
 - 12 $\zeta = \zeta + 1$, $p_{\text{RES}}^\zeta \rightarrow p_{\text{RES}}^{\zeta+1}$ and $p_{\text{SG}}^\zeta \rightarrow p_{\text{SG}}^{\zeta+1}$
 - 13 **end**
-

$H_\lambda = d\mathcal{F}_\lambda^2/dx_\lambda dx_\lambda^\top$ with respect to x_λ directly via built-in automatic differentiation in frameworks like TensorFlow or PyTorch. These derivatives provide first- and second-order information, including search directions and curvature estimates, to nonlinear nonconvex optimization algorithms like the PDIPM. The whole framework of the proposed method is shown in Fig. 3 and Algorithm 1.

IV. NUMERICAL RESULTS

The proposed method is tested on the modified NPCC system, which is a reduced model for the power grid of the northeastern United States and Canada. To evaluate the NPCC system's operation with substantial offshore integration, we incorporate four real large-scale offshore wind farms: South Fork Wind, located 35 miles east of Montauk Point, NY, is U.S.' first utility-scale offshore wind farm project, featuring 12 turbines and an offshore substation that generate 132 MW to serve 70,000 Long Island homes; Vineyard Wind 1, sited 35 miles off Massachusetts, comprises 62 turbines producing 800 MW-enough for over 400,000 homes and equivalent to removing 325,000 vehicles from the roads; Vineyard Wind 2 will deliver up to 1,200 MW to power 650,000 New England households; and the Vineyard Mid-Atlantic Offshore Wind project, in a lease area of the New York Bight, is projected to deploy 117 turbines to generate more than 2,000 MW for over 700,000 homes. Fig. 4 shows the actual geographic locations of the four offshore wind farms and the wind speed distribution in the surrounding Mid-Atlantic waters at a specific time.

The close proximity of these sites results in strong wind speed correlations, making simultaneous occurrences of both high and low wind speeds likely, which can easily lead to severe load-margin shortages. Furthermore, four transmission lines adjacent to the offshore wind farms have been equipped with DLR sensors. This project represents the first DLR demonstration tied to the U.S.'s first utility-scale offshore wind generation. High-fidelity sensors are deployed on critical transmission lines based on actual operational experience. The single-line diagram of the modified NPCC system with four offshore wind farms is shown in Fig. 5.

A. Multi-Fidelity DLR Fusion

Historical DLR data comprised hourly measurements over a one-year period from May 23, 2023 to May 24, 2024, totaling 8,807 observations. Although the data are collected as a continuous time series, the proposed MFDSPP model performs multi-fidelity DLR fusion in a snapshot-based manner, using low-fidelity and high-fidelity measurements at each individual time instant without exploiting temporal dependencies across time. Therefore, the samples were randomly shuffled and split into training (80%), testing (10%), and calibration (10%) sets for model development and evaluation. This random shuffling mitigates temporal dependence and ensures that the calibration and test samples are drawn from the same underlying operational distribution, making the exchangeability assumption required by CP approximately valid. In contrast, splitting the data strictly according to time order may introduce systematic distribution differences between the data subsets, which could confound snapshot-based learning and degrade the validity of uncertainty calibration. Additionally, we randomly introduced 5% missing values throughout the dataset to simulate sensor failures and communication packet loss. For a 20-mile transmission line with 50 spans, eight of which are equipped with direct sensors, we evaluated MFDSPP performance using those measurements. The proposed MFDSPP is evaluated against deterministic methods such as DNN, DBN, and MFNN, and probabilistic approaches such as DGP, DKL, DSPP, and MFDKL, using MAE, RMSE, and R^2 as metrics. The results are presented in Table I. It should be noted that DNN, DBN, DGP, DKL, and DSPP process and evaluate low-fidelity and high-fidelity inputs together without distinction, whereas MFNN, MFDKL, and MFDSPP employ a two-level multi-fidelity architecture similar to Fig. 2. Table I shows that

TABLE I
COMPARISON OF DIFFERENT DLR ESTIMATION MODELS

Type	Model	MAE	RMSE	R^2
Deterministic model	DNN	7.1984	8.9178	0.9982
	DBN	5.8920	7.2666	0.9988
	MFNN	4.7602	6.6343	0.9990
	DGP	21.0397	27.0324	0.9833
	DKL	7.9892	10.6479	0.9973
Probabilistic model	DSPP	3.4110	4.2832	0.9996
	MFDKL	3.6332	5.1045	0.9994
	MFDSPP	2.8336	3.8554	0.9997

multi-fidelity models consistently outperform direct evaluation

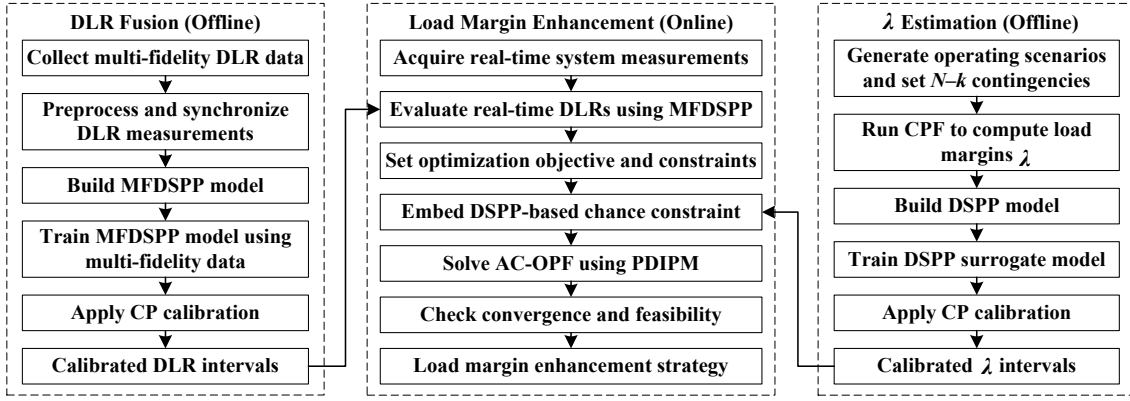


Fig. 3. Flowchart of the proposed load margin enhancement method considering multi-fidelity DLR fusion.

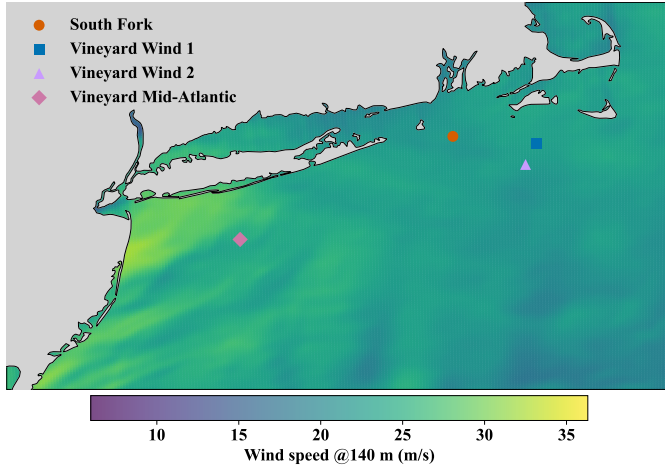


Fig. 4. Actual geographic locations of the four offshore wind farms and wind speed distribution in the surrounding Mid-Atlantic waters at a specific time.

models. In deterministic algorithms, MFNN yields better metrics than DNN and DBN. In probabilistic methods, MFDKL and MFDSPP achieve superior performance compared to DGP, DKL, and DSPP. Among all models, MFDSPP delivers the best performance. Its two-layer multi-fidelity architecture and weighted Gaussian-mixture approximation enable an MAE of 2.8336, significantly outperforming the next best model's 3.4110.

Beyond the point-estimate performance metrics reported in Table I, MFDSPP also provides quantifiable uncertainty intervals around its estimations. For the continuous 200-hour period from March 11 to March 20, 2024, we compared the MFDSPP $\pm 2\sigma$ uncertainty intervals and the CP calibrated $\pm q\sigma$ intervals against the true values, as shown in Fig. 6. In Fig. 6, the blue region represents the uncalibrated interval with a width of 123.37 MW, and the CP calibrated interval shown in orange has a width of only 15.29 MW, making it difficult to distinguish on the plot. The proportion of true values contained within these intervals decreased from 100% to 97.7% after calibration, approaching the theoretical rate of 95.45%. By applying CP, the intervals avoid excessive tightening, which could lead to RES curtailment and economic losses, while still ensuring a high probability of safe operation.

To examine the sensitivity of the proposed MFDSPP framework to data quality and sensor availability, Fig. 7 shows the MAE under varying missing-data rates and numbers of high-fidelity sensors. As shown in Fig. 7, for a fixed noise level, increasing the number of high-fidelity sensors consistently leads to a reduction in MAE, indicating that additional high-quality measurements enhance the model's ability to estimate DLRs accurately. This improvement is most pronounced when the sensor count increases from low to moderate levels, whereas further increases yield progressively smaller gains, suggesting diminishing returns beyond a certain sensor density. Conversely, for a fixed number of sensors, higher missing-data rates result in increased MAE, reflecting the reduced informational content available for model inference under degraded data quality. Importantly, this adverse effect is partially alleviated as more high-fidelity sensors are deployed, highlighting the complementary role of sensor redundancy in improving robustness against missing or noisy data. Overall, the results in Fig. 7 reveal a clear trade-off between sensor deployment density and data quality, offering practical guidance for system planners to balance monitoring cost, data reliability, and estimation accuracy. In addition, the average training time of the MFDSPP is approximately 17 minutes, which meets the requirement for rapid updating and retraining.

B. Load Margin Estimation

The active power outputs of SGs p_{SG} and the active power demands of loads p_d vary between 70% and 130%, while wind farm output p_{RES} ranges from 0% to 100%. The load is assumed to ultimately grow to 200% of the base load. The contingency set comprises 119 cases, including 14 $N-1$ generator outages, 54 $N-1$ line outages, and 50 $N-2$ double device outages. The $N-2$ contingencies explicitly include generator-generator, generator-line, and line-line outage combinations to comprehensively capture the impact of multiple simultaneous component failures on load margin estimation. The CPF-based data generation is performed entirely offline and does not affect real-time operation. In practice, to meet scalability requirements, the contingency set can be screened to include only critical $N-k$ events, and CPF computations are embarrassingly parallel and can be efficiently distributed

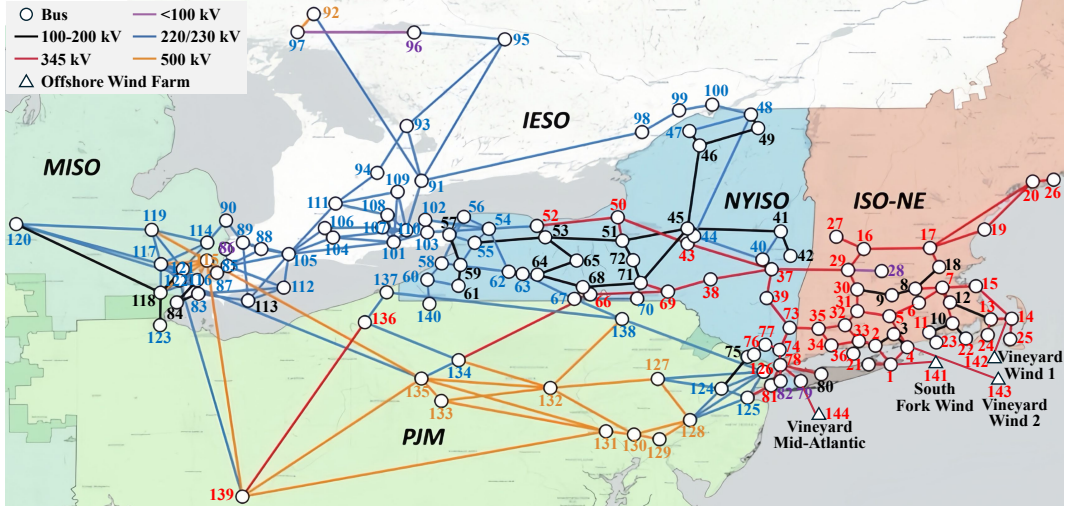


Fig. 5. One line diagram of modified NPCC system with four offshore wind farms.

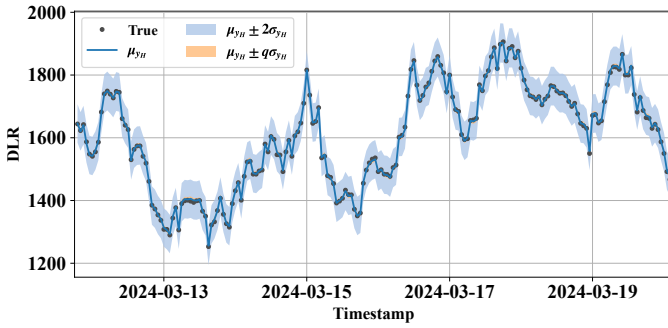


Fig. 6. MFDSP-based time-series DLR estimation results with CP calibrated uncertainty intervals.

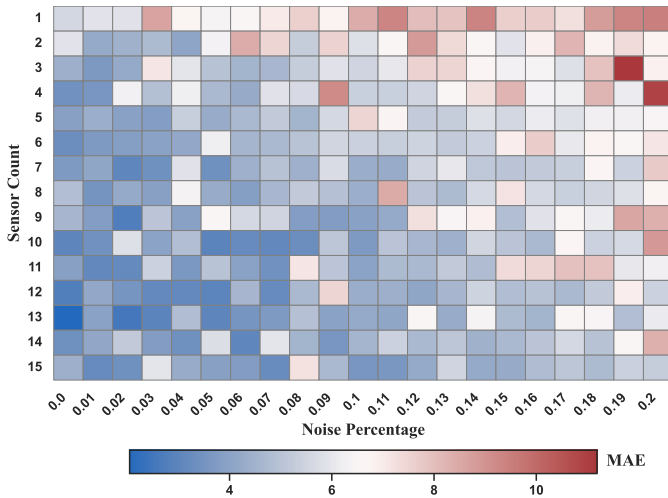


Fig. 7. Heatmap of MFDSP MAE across noise percentages and the number of high-fidelity sensors.

across computing resources. We generated 10,000 random operating scenarios and ran CPF analyses for each contingency to produce the samples used to train the DSPP model. The DSPP is compared against deterministic methods (DNN, DBN) and probabilistic approaches (DGP, DKL), with MAE, RMSE, and

R^2 results shown in Table II. Table II shows that the DSPP

TABLE II
COMPARISON OF DIFFERENT LOAD MARGIN ESTIMATION MODELS

Type	Model	MAE	RMSE	R^2
Deterministic model	DNN	0.0191	0.0291	0.9340
	DBN	0.0178	0.0283	0.9364
	DGP	0.0937	0.1134	-0.0307
Probabilistic model	DKL	0.0480	0.0680	0.6338
	DSPP	0.0105	0.0250	0.9504

outperforms all other deterministic and probabilistic models on every metric, achieving an MAE of 0.0105, which corresponds to a maximum deviation of 1.05% of the system's base load.

Additionally, Fig. 8 presents the DSPP's load margin estimates across 200 random scenarios, alongside the raw uncertainty intervals and those calibrated via CP. The blue

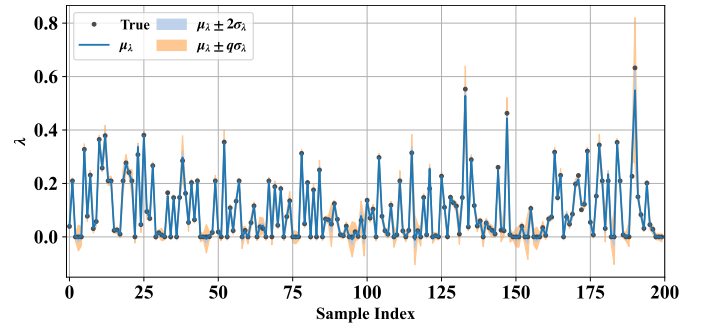


Fig. 8. DSPP-based load margin estimation results with CP calibrated uncertainty intervals.

shaded region represents the uncalibrated interval, which spans 0.0144, while the CP-adjusted interval in orange spans 0.0490. The coverage of true values rises from 58.35% to 97.0% after calibration, closely matching the 95.45% theoretical expectation. By widening the prediction band, CP mitigates the safety risks associated with insufficient margins. Fig. 6 and 8 show that CP can both tighten and widen the uncertainty

intervals, striking a balance between interval compactness and coverage.

It should be noted that under normal operating conditions, with unchanged network topology and contingency sets, the DSPP model does not require frequent retraining. In practice, retraining is typically needed only infrequently, on the order of months (e.g., quarterly to semi-annually), and is primarily triggered by structural topology changes that shift the operating regime beyond the modeled contingency set, such as the addition of new transmission lines or major generation units. In these cases, offline retraining of the DSPP surrogate and recalibration of the CP scores are required. The DSPP training time is approximately 420 seconds, which allows for relatively fast retraining in practice. Moreover, existing transfer learning techniques could further accelerate this process, although they are beyond the scope of this work [34].

C. Real-Time Load Margin Enhancement

For a continuous 24-hour scenario, hourly real-time load margin enhancement control is performed. At each hourly time step, an ACOF problem is solved subject to a comprehensive set of operational constraints. Specifically, the transmission line capacity constraints are enforced using either real SLR or DLR limits. Bus voltage magnitude constraints are imposed at every bus using standard operational ranges consistent with NPCC practice. The active and reactive power outputs of SGs are bounded by their respective minimum and maximum limits as specified in the NPCC system data. For RESs, the available active power at each time step is determined by the hourly wind power availability of each offshore wind farm. The corresponding reactive power capability is constrained by the inverter apparent power rating at the point of interconnection, modeled via the inverter capability circle, together with any applicable reactive power limits. The MFDSPP evaluates DLRs using real-time measurements to formulate transmission line capacity constraints, and the DSPP surrogate model is embedded within the PDIPM to enable rapid load margin assessment. All constraints in (2) - (8) and (12) are implemented as hard operational bounds in the hourly ACOF to ensure secure and consistent system operation and to enable a fair comparison between different control strategies. Since both the MFDSPP and DSPP models are trained using discretely sampled data drawn from the annual operational distribution, they are not sensitive to specific seasons or times. Consequently, the proposed load-margin enhancement method is applicable at any time of the year, provided that the operating conditions follow the same underlying operational distribution as the training data. In our implementation, the average computation time per hourly optimization is approximately 50 seconds, which is well within the one-hour update interval and thus suitable for real-time operation in this setting.

Fig. 9 shows the 24-hour active power flows on four key lines equipped with DLR sensors: (a) Line 2–33, (b) Line 4–5, (c) Line 12–13, and (d) Line 78–79. For most hours, the DLR-based power flow p_l (solid green lines) exceeds the SLR-based flow p_l (solid blue lines), as the DLRs are higher than the conservative SLRs. This effect is particularly evident

during periods of high wind generation between 15:00 and 23:00, when SLR-based operation repeatedly clips power flows at static limits, leading to unnecessary congestion and RES curtailment. By contrast, DLR-based control exploits available thermal headroom and enables higher power transfers. At the same time, Fig. 9 also demonstrates the protective role of DLR under adverse weather conditions. Specifically, at 17:00 on Line 2–33 (subplot (a)), 01:00 on Line 4–5 (subplot (b)), and 18:00 on Line 12–13 subplot (c)), the DLR temporarily falls below the SLR, imposing more restrictive limits to safeguard conductor integrity. These cases highlight that DLR not only increases transmission utilization when conditions permit, but also enforces stricter constraints when necessary, whereas exclusive reliance on SLR could expose lines to thermal overstressing and potential safety risks.

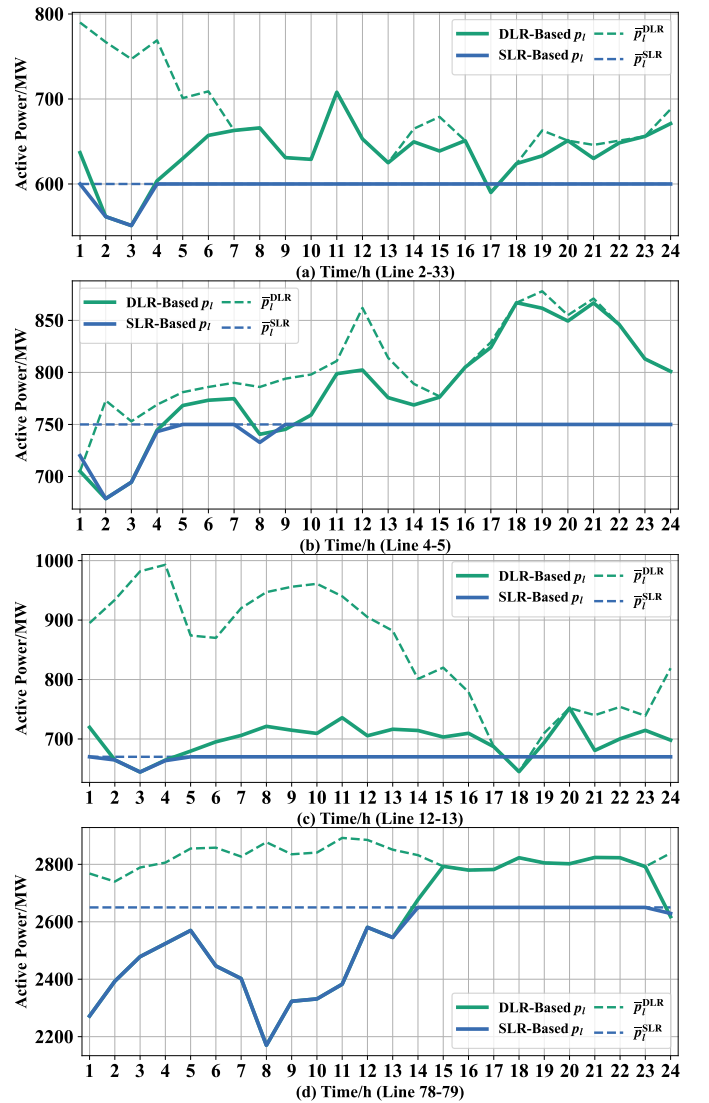


Fig. 9. Comparison of 24-hour transmission line active power flows under DLR and SLR. (a) Line 2–33, (b) Line 4–5, (c) Line 12–13, and (d) Line 78–79 illustrate key transmission lines equipped with DLR sensors. Solid lines denote actual power flows under DLR- and SLR-based control, while dashed lines indicate the corresponding line rating limits.

Correspondingly, Fig. 10 illustrates the 24-hour active power outputs of four offshore wind farms under DLR- and SLR-

based control. Subplots (a)–(d) correspond to the South Fork Wind, Vineyard Wind 1, Vineyard Wind 2, and Vineyard Mid-Atlantic wind farms, respectively. Across all four cases, the DLR-based RES power output p_{RES} (solid green lines) follows the forecasted generation $\hat{p}_{\text{RES}}$ (black dashed lines) more closely than the SLR-based output p_{RES} (solid blue lines). This behavior is particularly evident during periods of high wind availability, when conservative SLR constraints limit generation and lead to RES curtailment. In contrast, DLR-based capacity constraints relax these bottlenecks and enable higher RES delivery. Over the full 24-hour horizon, the DLR-based control unlocks an additional 2,189.7 MWh of clean energy compared to the SLR-based case.

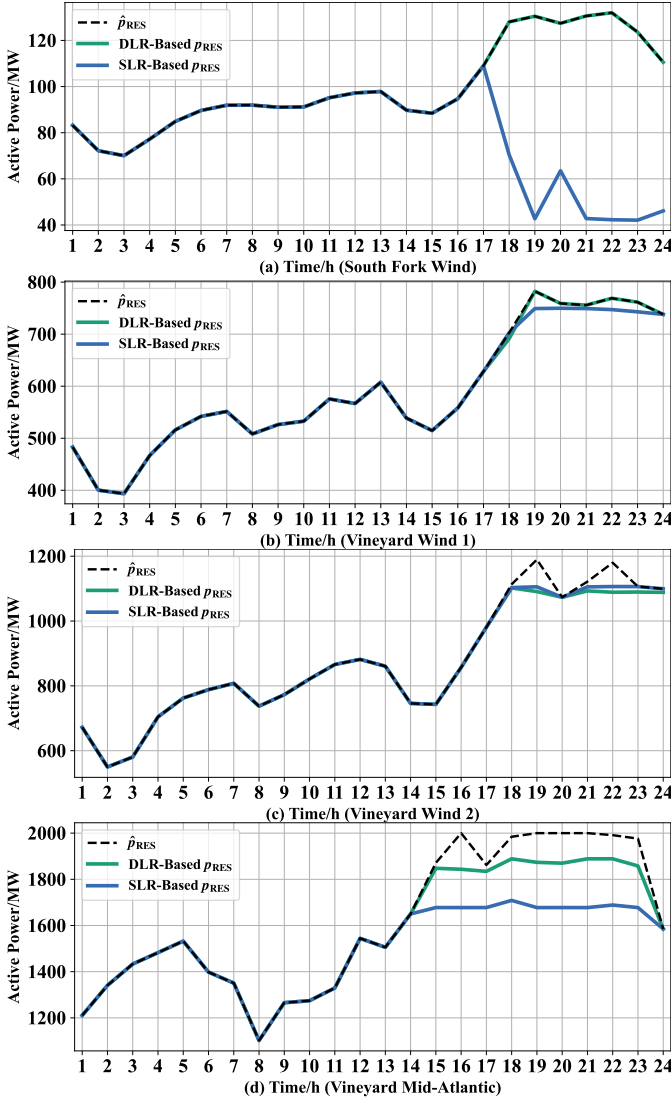


Fig. 10. Comparison of 24-hour offshore wind farm active power outputs under DLR and SLR. (a) South Fork Wind, (b) Vineyard Wind 1, (c) Vineyard Wind 2, and (d) Vineyard Mid-Atlantic illustrate offshore wind farms. Solid lines denote actual renewable power outputs under DLR- and SLR-based control, while dashed black lines indicate forecasted generation.

Fig. 11 presents a 24-hour comparison of load margins λ under different real-time control schemes, together with a hard safety threshold $\underline{\lambda} = 0.10$ (10% of the base load). This threshold is selected to represent a reasonably conservative operating

buffer and to demonstrate the feasibility of the proposed method. This threshold is treated as a user-defined parameter that can be adjusted to reflect different system characteristics or operational requirements, and alternative values can be readily accommodated within the same framework. When load margin constraints are omitted, the resulting load margin drops to as low as 0.0616 (black dashed line), indicating that voltage stability margins can be severely eroded during real-time operation. Such narrow margins would be rapidly consumed by normal demand fluctuations, underscoring the necessity of explicitly enforcing load margin constraints. When load margin constraints are imposed, all post-control load margins remain above the safety threshold, as shown by the SLR-based (solid blue line) and DLR-based (solid green line) results. The chance constraint further introduces a probabilistic safety buffer by enforcing $\mu_{\lambda} - q_{\lambda}\sigma_{\lambda} \geq 0.1$. The zoomed-in inset in the upper-right corner highlights the low-margin region and shows that, under the combined DLR and chance-constrained control, the CPF-based validation results (solid purple line) achieve a minimum load margin of 0.105 over the 24-hour horizon. This confirms that the integration of chance constraints and CP calibration provides a robust and quantifiable probabilistic safeguard against voltage instability in real-time operation.

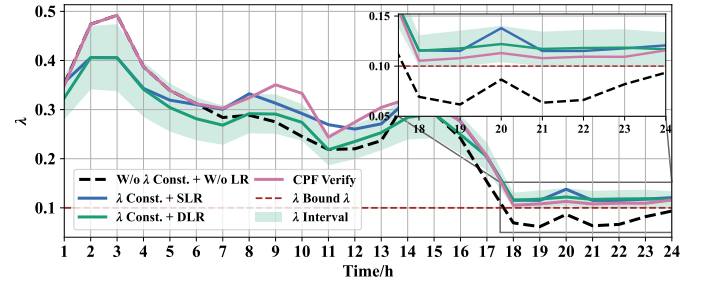


Fig. 11. Comparison of 24-hour load margins under different real-time control schemes. The dashed red line denotes the safety threshold $\underline{\lambda} = 0.10$. The inset provides a zoomed-in view of the low-margin region, illustrating the effectiveness of chance constraints and CPF-based validation in maintaining adequate voltage stability margins.

Fig. 12 presents the 24-hour operating cost curve. Over this period, the unconstrained baseline (no load margin constraint or line rating enforcement) incurs \$4,353,027 (black dashed line). Introducing SLRs with load margin constraints raises the cost to \$4,485,297 (solid blue line). By comparison, DLR-based control results in \$4,382,677 (solid green line), approximately 2.34% below the SLR case, by unlocking extra transfer capacity, reducing curtailment, and capturing more low-cost wind energy. While these savings do not explicitly account for the capital costs of installing DLR sensors or the computational infrastructure required for the DSPP/MFDSPP framework, they represent operational benefits that can be reinvested in deploying and advancing DLR technology and in enhancing operators' situational awareness of transmission lines.

Furthermore, Fig. 13 examines the relationship between the normalized aggregate wind power output p_{RES} and the load margin λ under unconstrained operation, i.e., without explicit load margin constraints or line rating enforcement.

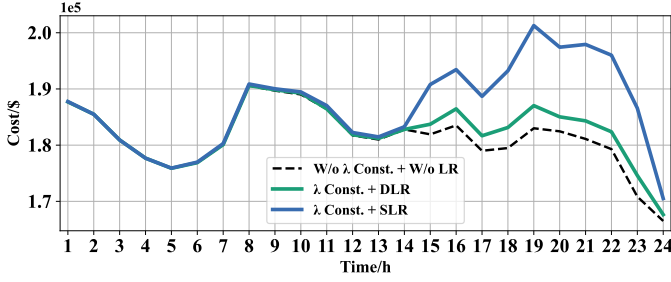


Fig. 12. Comparison of 24-hour operating costs under different control schemes.

Fig. 13 (a) shows the 24-hour time-series comparison, where the two variables exhibit opposite trends: periods of high wind generation correspond to reduced load margins. Fig. 13 (b) further quantifies this relationship using a scatter plot of hourly data points and a least-squares fit. The fitted model yields $\lambda = -0.798 p_{\text{RES}} + 0.446$ with a Pearson correlation coefficient of $r = -0.90$, confirming a strong inverse correlation between wind power output and load margin. These results indicate that, without appropriate security constraints, increasing renewable generation can significantly erode voltage stability margins, thereby underscoring the necessity of incorporating load margin constraints and accurate DLR limits in real-time scheduling and control as RES penetration increases.

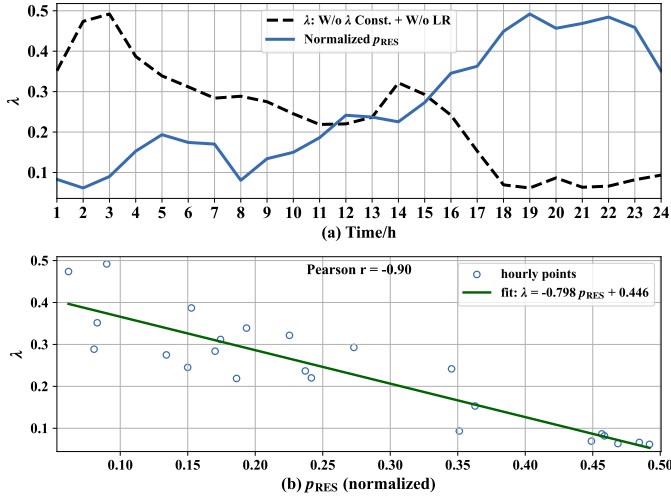


Fig. 13. Correlation analysis between normalized aggregate wind power output p_{RES} and load margin λ under unconstrained operation. (a) 24-hour time-series comparison showing the opposing trends of wind power output and load margin. (b) Scatter plot of hourly data with least-squares regression, illustrating the strong inverse correlation between p_{RES} and λ .

D. Comparison With the Other Methods

In power system optimization and control, sensitivity-based approaches are widely adopted to guide control actions by approximating the gradient of security metrics with respect to decision variables. To evaluate the effectiveness of the proposed method, the following benchmarks are considered:

- **Classical CPF-based sensitivity method + DLR**

Sensitivities of the load margin λ with respect to control

variables are estimated via finite-difference approximations of the CPF solution, e.g.,

$$\frac{\partial \lambda}{\partial x_{\lambda,k}} \approx \frac{\lambda(x_{\lambda} + \Delta x_{\lambda,k} e_{\lambda,k}) - \lambda(x_{\lambda} - \Delta x_{\lambda,k} e_{\lambda,k})}{2\Delta x_{\lambda,k}}$$

$$e_{\lambda,k} = [0, \dots, 0, 1, 0, \dots, 0]^T, \quad k = 1, \dots, n \quad (31)$$

This method represents a classical linear sensitivity factor approach grounded in physical modeling. At each control interval, the load margin λ sensitivities with respect to all control variables (p_{RES} and p_{SG}) are computed by perturbing each component of the control vector and evaluating the corresponding CPF solutions under all $119 N - k$ contingencies. Based on the resulting sensitivity information, a candidate control update is generated using a first-order approximation of the load margin. The updated operating point is then verified using CPF to ensure that the resulting load margin satisfies the prescribed requirement. If the requirement is not met, the control update is adjusted (e.g., by reducing the step size) and re-validated using CPF.

- **DBN-based surrogate method + DLR**

A DBN-based surrogate is trained to approximate the load margin λ . Jacobian and Hessian of the load margin λ with respect to control variables are obtained either via finite-difference approximation or directly via built-in automatic-differentiation functions. While this approach is computationally efficient, its point predictions may suffer from approximation errors, particularly in low-margin regions.

- **DSPP-based surrogate without CP + DLR**

The DSPP surrogate provides probabilistic predictions of the load margin, but uncertainty intervals are directly derived from the model without calibration. As a result, according to Fig. 8, miscalibration can lead to insufficient safety margins in certain operating conditions.

- **Proposed DSPP-based surrogate with CP + DLR**

The proposed approach combines DSPP with CP to calibrate uncertainty intervals and enforce chance-constrained load margin λ requirements in real-time.

To ensure fair comparison, all control actions generated by the above methods are post-validated using CPF. Specifically, the control decisions are applied as inputs to CPF, and the resulting true load margins are reported. The 24-hour CPF-verified comparison results are shown in Fig. 14. It can be seen that the classical CPF-based sensitivity method can strictly satisfy the load margin requirements through a large number of CPF evaluations. However, its computational burden is prohibitive: in the studied system, a single control interval requires approximately 38 hours due to repeated CPF evaluations over all control variables and $N - k$ contingencies. As a result, this approach is entirely incompatible with hourly operation, let alone faster control cycles such as 15-minute or 5-minute intervals. In contrast, the remaining three methods can all be executed within approximately 50 seconds per control interval, corresponding to a speedup of more than 2,000 $\times$ compared with the classical CPF-based sensitivity approach, and are therefore suitable for real-time implementation. Among these

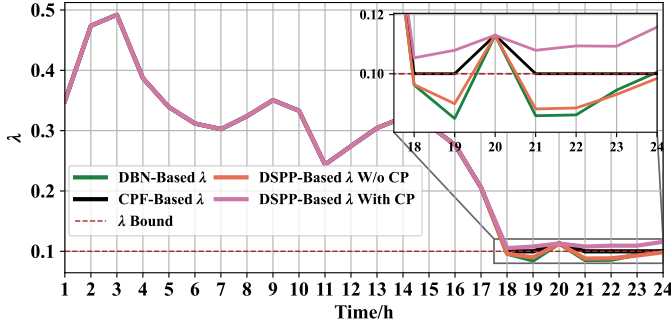


Fig. 14. Comparison of 24-hour load margin λ under different control methods. The inset highlights robustness differences in the low-margin region among the compared methods.

real-time-capable methods, the DBN-based approach suffers from estimation errors in λ , leading to violations of the 0.1 margin threshold at certain times. The DSPP-based method without CP also fails to fully meet the requirement because of errors in its estimated λ uncertainty intervals. As highlighted in the zoomed-in inset of the low-margin region, only the proposed DSPP-based method with CP maintains the load margin above the prescribed threshold throughout the entire 24-hour horizon. These results demonstrate that accurate uncertainty quantification, together with CP-based calibration, is essential for achieving reliable load margin-aware real-time control.

E. Practical Implications and Generalization

The proposed multi-fidelity DLR fusion and load margin-aware control framework provides tangible benefits for power system operation under high RES penetration. For system operators, integrating calibrated DLR constraints with probabilistic load margin constraints enables real-time dispatch decisions that balance transmission utilization and voltage stability. Numerical results demonstrate that the framework consistently maintains load margins above 10% of the base load while allowing higher power transfers under favorable conditions, thereby improving operational security and reducing the risk of voltage instability during periods of high wind generation and heavy loading. Moreover, the framework enhances operational efficiency by reducing RES curtailment. In the NPCC case study, DLR-based control unlocks an additional 2,189.7 MWh of clean energy and reduces operating costs by approximately 2.34% compared to conventional SLR-based operation. These benefits are achieved through more effective utilization of existing transmission capacity rather than physical network expansion, making the approach well suited for near-term deployment.

For system planners, the proposed approach quantifies the relationship between sensor deployment, data quality, and operational performance. The sensitivity results reveal diminishing gains in estimation accuracy beyond a moderate number of high-fidelity sensors, offering practical guidance for designing cost-effective monitoring schemes. Based on these results, the proposed DLR framework can be readily transferred and adopted in other systems with similar monitoring and operational requirements.

For policymakers, the results demonstrate how grid-enhancing technologies such as DLR, when integrated with data-driven and uncertainty-aware control, can enable higher levels of RES integration while preserving system reliability. The quantified improvements in RES utilization and operating efficiency provide an empirical basis for developing policies that support the deployment of advanced sensing, data fusion, and control technologies in transmission systems.

Although the case studies focus on the NPCC system with large-scale offshore wind integration, the proposed framework is broadly applicable to other power systems, grid topologies, and operating configurations. The multi-fidelity DLR fusion approach is not tied to a specific sensor type or network structure and can readily incorporate different combinations of meteorological data and direct line measurements. Similarly, the DSPP-based surrogate modeling and CP calibration are data-driven and can be retrained for systems with different sizes, contingency sets, generation mixes, and load characteristics.

From a modeling perspective, the proposed framework is not tied to a specific system configuration. Changes in SG portfolios, the presence of large electric loads, or buses with co-located load and generation are naturally reflected through the system operating conditions used in AC power flow and CPF analyses. Such variations influence the input features and training data of the surrogate model, but do not alter the structure of the proposed framework or its underlying control formulation.

The applicability of the method is also not limited to offshore wind-dominated systems. It can be extended to systems with onshore wind, solar photovoltaics, or hybrid renewable energy mixes, provided that representative operational data are available for surrogate training and calibration. As renewable penetration increases and system uncertainty becomes more pronounced, the proposed uncertainty-aware, load margin-constrained control paradigm offers a scalable and flexible foundation for enhancing both operational efficiency and reliability across diverse power systems and operating conditions.

V. CONCLUSION

This paper proposes a multi-fidelity DLR fusion-based load margin enhancement method, accounting for large-scale offshore wind farm integration. The proposed MFDSP can fuse multi-fidelity DLR data from diverse sensors to estimate the minimum line rating and quantify its uncertainty, while the DSPP can replace time-consuming CPF calculations to rapidly assess load margins under a large number of $N - k$ contingencies. CP is successfully applied to both MFDSP and DSPP to correct their uncertainty intervals. The DLR-based line capacity constraints and DSPP-derived load-margin constraints are integrated into the real-time control framework to optimally manage offshore wind farm output. Case studies on the NPCC system with four large-scale offshore wind farms validate the effectiveness of the proposed method, showing that it maintains a load margin greater than 10% of the base load, unlocks an additional 2,189.7 MWh of clean energy,

reduces operating costs by approximately 2.34%, and achieves more than a 2,000× speedup compared with classical CPF-based sensitivity methods. In future research, other grid-enhancing technologies, grid-forming inverter control, adaptive calibration under distribution drift, and techno-economic considerations of sensing infrastructure will be incorporated to further enhance its overall system-level effectiveness.

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